

Week 1 Data Exploration Report

Dataset: DePaul Dataset

Team: SLU 0203 DVT Team 3

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Introduction

This report presents the initial exploration and structural assessment of the DePaul dataset. The objective of this stage is to understand the dataset structure, evaluate its quality, and identify early patterns that will guide further analysis.

Dataset Overview and Documentation

The DePaul dataset contains records related to prospective students or applicants interacting with the DePaul University system. The data appears to originate from a **Customer Relationship Management (CRM) or admissions tracking system** used to manage student inquiries, applications, and communication history.

An overview of the dataset can be accessed here:
<https://docs.google.com/document/d/1b3BTouVvfmgq2Lk7BCYPb82wOTaEyzMvKPEdTkBWa-s/edit?usp=sharing>

Data Dictionary

A detailed data dictionary containing variable descriptions, data types, and example values can be accessed here:

<https://docs.google.com/spreadsheets/d/1GwSUEa-eEY6irJ3LkRjTuUame-ry88JA7kXpVHIgLCg/edit?usp=sharing>

Data Quality Report

This report presents an analysis of the applicant dataset, focusing on identifying data quality issues that may affect further analysis and modeling. The dataset contains information related to applicant details, communication attempts, outreach activities, and admission-related records.

As part of the data preparation process, a detailed data quality assessment was conducted to identify potential issues such as missing values, duplicate records, inconsistent data types, and possible outliers.

This report can be accessed here:

https://docs.google.com/document/d/1WfoSI8XvBPWssKJ0oRhTDuNH9Zp1qp_pWz4r6Fg0I8/edit?usp=sharing

Exploratory Data Analysis (EDA) Report

Exploratory Data Analysis (EDA) was conducted to gain deeper insights into the structure and patterns within the applicant dataset. The goal of this stage is to summarize key characteristics of the data, identify trends, and uncover relationships between variables that may be useful for further analysis.

Through EDA, various statistical summaries and visualizations are used to better understand the distribution of variables, detect potential anomalies, and explore how different features interact within the dataset. This process helps transform raw data into meaningful insights that support informed decision-making.

The detailed EDA findings, including visualizations and analytical observations, are documented in the report linked below.

Full EDA Report:

<https://docs.google.com/document/d/1yEOCg2gy01y0mCIZuHJNEOdukBSww2qTibra3dVe7tc/edit?usp=sharing>

